GLA University

Maths Assignment

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Q1 By the method of least squares, find the straight line that best fits the following data:

X	1	2	3	4	5
Y	14	27	40	55	68

Solution

X	Y	XY	X^2
1	14	14	1
2	27	54	4
3	40	120	9
4	55	220	16
5	68	340	25
15	204	748	55

$\Sigma y = na + b\Sigma x$

$204 = 5a + 15b$ — (1)

$\Sigma xy = a\Sigma x + b\Sigma x^2$

$748 = 15a + 55b$ — (2)

$748 = 15a + 55b$

$612 = 15a + 45b$

$\frac{136}{10} = b$

$\boxed{b = 13.6}$

$204 = 5a + 13.6 \times 15$

$\boxed{a = 0}$

Q2 Show that the line of fit to the following data is given by $y = 0.7x + 11.285$

x	0	5	10	15	20	25	75
y	12	15	17	22	24	30	120

Sol Eq^n of straight line to be fitted is

$y = 0.7x + 11.285$ —— (given)

Hence $n = 6$

Normal equation → $\Sigma y = na + b\Sigma x$

$\Sigma xy = a\Sigma x + b\Sigma x^2$

xy	0	75	170	330	480	750	1805
x^2	0	25	100	225	400	625	1375

$75\ (6a + 75b = 120)$ —— ①

$6\ (75a + 1375b = 1805)$ —— ②

$450a + 5625b = 9000$

$450a + 8250b = 10830$

$-2625b = -1830$

$\boxed{b = 0.7}$

$450a = 9000 - 3937.5$

$a = \frac{5062.5}{450}$

$\boxed{a = 11.285}$

So that the value: $y = a + bx$

$$\boxed{y = 11.285 + 0.7x}$$

Q3 The marks secured by recruits in the selection test (X) and the proficiency test (Y) are given below.

Serial No.	1	2	3	4	5	6	7	8	9
X	10	15	12	17	13	16	24	14	22
Y	30	42	45	46	33	34	40	35	39

Calculate the rank correlation co-efficient.

Solution →

S.No	X	Y	$R_1(X)$	$R_2(Y)$	$d = R_1(X) - R_2(Y)$	d^2
1	10	30	9	9	0	0
2	15	42	5	3	2	4
3	12	45	8	2	6	36
4	17	46	3	1	2	4
5	13	33	7	8	−1	1
6	16	34	4	7	−3	9
7	24	40	1	4	−3	9
8	14	35	6	6	0	0
9	22	39	2	5	−3	9
						72

$$\rho = 1 - \left[\frac{6\Sigma d^2}{n(n^2-1)}\right]$$

$$\rho = 1 - \left[\frac{6 \times 72}{9(9^2-1)}\right]$$

$$\rho = 1 - \left[\frac{432}{9(80)}\right]$$

$$\rho = 1 - \left[\frac{432}{720}\right]$$

$$\rho = 1 - 0.6$$

$$\boxed{\rho = 0.4}$$

Q4 A sample of 12 fathers and their eldest son gave the following data about their height in inches:-

father	65	63	67	64	68	62	70	66	68	67	69	71
son	68	66	68	65	69	66	68	65	71	67	68	70

Calculate the co-efficient of rank correlation.

Solution

father	son	R_1 (father)	R_2 (son)	$d = R_1 - R_2$	d^2
65	68	9	5.5	3.5	12.25
63	66	11	9	2	4
67	68	6.5	5.5	1	1
64	65	10	10.5	-1.5	2.25
68	69	4.5	3	1.5	2.25
62	66	12	10	2	4
70	68	2	5.5	-3.5	12.25
66	65	8	11.5	-3.5	12.25
68	71	4.5	1	-3.5	12.25
67	67	6.5	8	-2.5	6.25
69	68	3	5.5	-2.5	6.25
71	70	1	2	-1	1
					76

$n = 12$

$$r = 1 - \left[\frac{6 \Sigma d^2}{n(n^2-1)}\right]$$

$$r = 1 - \left[\frac{456}{12(12^2-1)}\right]$$

$$r = 1 - \left[\frac{456}{1716}\right]$$

$$r = 1 - 0.2657$$

$$\boxed{r = 0.73}$$

Q5 Find the coefficient of correlation b/w X & Y from the table of their values :-

X	1	3	4	6	8	9	11	14
Y	1	2	4	4	5	7	8	9

Solution

X	Y	XY	X^2	Y^2
1	1	1	1	1
3	2	6	9	4
4	4	16	16	16
[illegible]	4	16		16
6	5	24	36	25
8	7	40	64	49
9	8	63	81	64
11	9	88	121	81
14		126	196	
56	40	364	524	256

$$r_{xy} = \frac{n\Sigma xy - \Sigma x \Sigma y}{\sqrt{n\Sigma x^2 - (\Sigma x)^2}\sqrt{n\Sigma y^2 - (\Sigma y)^2}}$$

$n = 8$

$$r_{xy} = \frac{8 \times 364 - 56 \times 40}{\sqrt{8 \times 524 - (56)^2}\sqrt{8 \times 256 - (40)^2}}$$

$$r_{xy} = \frac{2912 - 2240}{\sqrt{1056}\sqrt{448}}$$

$$r_{xy} = \frac{672}{687.81}$$

$$r_{xy} = 0.977$$

06 Calculate the coefficient of correlation b/w x and y series from the following data.

$\Sigma (x-\bar{x})^2 = 136$, $\Sigma (y-\bar{y})^2 = 138$,

$\Sigma (x-\bar{x})(y-\bar{y}) = 122$

Solution

$$r(x,y) = \frac{\Sigma (x-\bar{x})(y-\bar{y})}{\sqrt{\Sigma (x-\bar{x})^2 \Sigma (y-\bar{y})^2}}$$

$$r(x,y) = \frac{122}{\sqrt{136 \times 138}}$$

$$\boxed{r(x,y) = 0.89}$$

07 The following result were obtained from marks in Applied Mechanics and Engineering Mathematics in an examination:

	Applied Mathematics (x)	Engineering Mathematics (y)
Mean	47.5	39.5
Standard deviation	16.8	10.8

$r = 0.95$

find both the regression equation. Also estimate the value of y for $X = 30$.

Sol

$\bar{x} = 47.5$, $\sigma_x = 16.8$

$\bar{y} = 39.5$, $\sigma_y = 10.8$

Line of regression of y on x :

$(y-\bar{y}) = b_{yx}(x-\bar{x})$

$y - 39.5 = b_{yx}(x-47.5)$

$$\boxed{y = 0.61x + 10.52}$$

$$b_{yx} = r\frac{\sigma_y}{\sigma_x}$$

$$\Rightarrow 0.95 \times \frac{10.8}{16.8}$$

$$\boxed{b_{yx} = 0.6}$$

Line of Regression x on y

$(x-\bar{x}) = b_{xy}(y-\bar{y})$ $\quad \therefore b_{xy} = 1.47$

$x - 47.5 = 1.47(y-39.5)$

$\boxed{x = 1.47y - 10.56}$ at $x=30$

$y = 0.61 \times 30 + 10.52$

$\boxed{y = 28.83}$

Q8 → In a study between the amount of rainfall and the quantity of air pollution removed the following data were collected.

Daily rainfall (in cm)	4.3	4.5	5.9	5.6	6.1	5.2	3.8	2.1
Pollution removed (mg/m^3)	12.6	12.1	11.6	11.8	11.4	11.8	13.2	14.1

find the regression line of y on x.

Solution

Daily rainfall (x)	Pollution removed (y)	xy	y^2	x^2
4.3	12.6	54.18	158.76	18.49
4.5	12.1	54.45	146.41	20.25
5.9	11.6	68.44	134.56	34.81
5.6	11.8	66.08	139.24	31.36
6.1	11.4	69.54	129.96	37.21
5.2	11.8	61.36	139.24	27.04
3.8	13.2	50.16	174.24	14.44
2.1	14.1	29.61	198.81	4.41
37.5	98.6	453.82	1221.22	188.01

Regression line y on x.

$$(y-\bar{y}) = b_{yx}(x-\bar{x})$$

$$b_{yx} = \frac{n\Sigma xy - \Sigma x \Sigma y}{n\Sigma x^2 - (\Sigma x)^2}$$

$$= \frac{8 \times 453.82 - 37.5 \times 98.6}{8 \times 188.01 - (37.5)^2}$$

$$= \frac{3630.56 - 3697.5}{1504.08 - 1406.25}$$

$$\Rightarrow \frac{-66.94}{97.83}$$

$$\boxed{b_{yx} \Rightarrow -0.68}$$

Q9 Two lines of regression is given by.

$x + 2y - 5 = 0$ and $2x + 3y - 8 = 0$ variance of $x = 12$

Calculate:

(a) the mean value of x and y (b) variance of y

(c) Correlation coefficient $\rho(x, y)$

Sol Given that

(a)

$$x + 2y = 5 \quad -(1)$$

$$2x + 3y = 8 \quad -(2)$$

$$x = 5 - 2y \quad -(3)$$

Put in (2)

$$10 - 4y + 3y = 8$$

$$\boxed{y = 2}$$

Put in (3)

$$x = 5 - 2 \times 2$$

$$\boxed{x = 1}$$

As we know that regression line passes through $(\bar{x}, \bar{y})$

$$\boxed{\bar{x} = 1, \ \bar{y} = 2}$$

given that.

(ii) regression line

$x = -2y + 5$ — (4) (x on y)

$y = \frac{-2}{3}x + \frac{8}{3}$ — (5) (y on x)

from (4) & (5)

$b_{xy} = r\frac{\sigma_x}{\sigma_y} = -2$ — (6)

$b_{yx} = r\frac{\sigma_y}{\sigma_x} = \frac{-2}{3}$ — (7)

We have

$r^2 = b_{yx} \cdot b_{xy}$

$r^2 = -2 \times \frac{-2}{3} = \frac{4}{3}$

$r = \sqrt{\frac{4}{3}} = \pm 1.15$ (which is not possible)

So, regression lines

$y = \frac{-x}{2} + \frac{5}{2}$ (y on x) — (8)

$x = \frac{-3}{2}y + 4$ (x on y) — (9)

from (8) & (9)

$b_{yx} = r\frac{\sigma_y}{\sigma_x} = \frac{-1}{2}$ — (10)

$b_{xy} = r\frac{\sigma_x}{\sigma_y} = \frac{-3}{2}$ — (11)

$r^2 = \frac{-1}{2} \times \frac{-3}{2} = \frac{3}{4} = 0.86$

b_{xy}, b_{yx} and r having same sign

given that,

$\sigma_x^2 = 12$; $\sigma_x = \sqrt{12}$

from Eq (10)

$$-0.86 \times \frac{\sigma_y}{\sigma_x} = \frac{0.86}{\sqrt{12}} \times \sigma_y = -\frac{1}{2}$$

$$\sigma_y = -\frac{1\sqrt{12}}{2(-0.86} = \frac{\pm 3.46}{\pm 1.72}$$

$\sigma_y = 2.014$

$$\boxed{\sigma_y^2 = 4.056}$$

Q10 An analyst for a company was studying travelling expenses(y) in rupees and duration (x) of these trips for 102 sales trip. He has found relation b/w x and y linear and data as follow.

$\Sigma x = 510$, $\Sigma y = 7140$, $\Sigma x^2 = 4150$, $\Sigma xy = 54900$

$\Sigma y^2 = 740200$

Calculate: (i) regression lines 2.

(ii) A given trip has to take 7 days. How much money should be allowed so that they will not run shot of money?

Sol $n = 120$

$$b_{xy} = \frac{n\Sigma xy - \Sigma x \Sigma y}{n\Sigma y^2 - (\Sigma y)^2} \Rightarrow \frac{120 \times 54900 - 510 \times 7140}{120 \times 740200 - (7140)^2}$$

$$b_{xy} = \frac{6588000 - 3641400}{88824000 - 50979600} \Rightarrow \frac{2946600}{37844400} \Rightarrow 0.07$$

$$\boxed{b_{xy} = 0.077}$$

$$b_{yx} = \frac{n\Sigma xy - \Sigma x \Sigma y}{n\Sigma x^2 - (\Sigma x)^2}$$

$$b_{yx} = \frac{120 \times 54900 - 510 \times 7140}{120 \times 4150 - (510)^2}$$

$$b_{yx} = \frac{2946600}{237900} \Rightarrow 12.38$$

$$\boxed{b_{yx} = 12.38}$$

$$b_{yx} = \frac{n\Sigma xy - \Sigma x \Sigma y}{n\Sigma x^2 - (\Sigma x)^2}$$

$$= \frac{120 \times 54900 - 510 \times 7140}{120 \times 4150 - (510)^2}$$

$$\Rightarrow \frac{2946600}{498000 - 260100} \Rightarrow \frac{2946600}{237900}$$

$$\boxed{b_{yx} = 12.38}$$

Q11. fit a binomial distribution of the following frequency data.

X	0	1	2	3	4
f	30	62	46	10	2

Solution Binomial distribution of above frequency distribution =

$N(p+q)^n$.

N = total frequency

n = no. of observation − 1

$$\text{mean} = np = \frac{\Sigma f(x)}{\Sigma f}$$

x	f	f(x)
0	30	0
1	62	62
2	46	92
3	10	30
4	2	8
	150	192

$$\text{Mean} = \frac{192}{150} = 1.28$$

$np = 1.28$

$n = 5 - 1 = 4$

$p = \frac{1.28}{4} = \boxed{0.32}$

but,

$p + q = 1$, $q = 1 - 0.32$

$\boxed{q = 0.68}$

So, binomial distribution =

$\Rightarrow 150(0.32 + 0.68)^4$

$\Rightarrow \boxed{150}$ Ans

Q12

(a) A binomial variable X satisfies the relation $9P(X=4) = P(X=2)$. When $n=6$, find the value of the parameter p and $P(X=1)$

Sol

We have,

$P(r) = {}^{n}C_r \, p^r q^{n-r}$

$P(4) = {}^{6}C_4 \, p^4 q^2$ — (1)

$P(2) = {}^{6}C_2 \, p^2 q^4$ — (2)

given that

$9P(4) = P(2)$

$9 \cdot {}^{6}C_4 \, p^4 q^2 = {}^{6}C_2 \, p^2 q^4$

$9p^4q^2 - p^2q^4 = 0$

$p^2q^2(9p^2 - q^2) = 0$

$p^2q^2 \neq 0$

$9p^2 - q^2 = 0$ — (3)

But, $p+q=1$, $p = 1-q$, $q = 1-p$

$9p^2 - (1-p)^2 = 0$

$9p^2 - (1 + p^2 - 2p) = 0$

$8p^2 + 2p - 1 = 0$

$8p^2 + 4p - 2p - 1 = 0$

$4p(2p+1) - 1(2p+1) = 0$

$(2p+1)(4p-1)$

$p = -\frac{1}{2}$ ✗ this is not possible

$p = \frac{1}{4}$ ✓

$p = \frac{1}{4}$

$q = 1 - \frac{1}{4} = \boxed{\frac{3}{4}}$

$P(1) = {}^{6}C_1 \left(\frac{1}{4}\right)^1 \left(\frac{3}{4}\right)^5$

$= {}^{3}\cancel{6} \times \frac{1}{\cancel{4}_2} \times \left(\frac{3}{4}\right)^5$

$\boxed{P(1) = 0.355}$

(b) In 800 families with 5 children each, how many families would expected to have

(i) 3 boys and 2 girls, (ii) 2 boys and 3 girls (iii) No girls (iv) atmost two girls (Assume boys and girls are equal)

Sol let the probability of boy → p

and probability of girl = q

$p = \frac{1}{2}$, $q = \frac{1}{2}$.

$n = 5$, $N = 800$

(i) Prob. of 3 boys & 2 girls.

$P(X=3) = {}^5C_3 \left(\frac{1}{2}\right)^3 \left(\frac{1}{2}\right)^2$

$= 10 \times 0.25 \times 0.125$

$= 0.312$

No. of families having three boys and two girls

$\Rightarrow 800 \times 0.312 \Rightarrow 250$ Ans

(ii) 2 boys and three girls,

$P(X=2) = {}^5C_2 \left(\frac{1}{2}\right)^2 \left(\frac{1}{2}\right)^3$

$\Rightarrow 10 \times 0.25 \times 0.0125$

$\Rightarrow 0.312$

No. of families having three girls two boys

$\Rightarrow 800 \times 0.312 = 250$ Ans

(iii) No girls = all are boys

$P(5) = {}^5C_5 \left(\frac{1}{2}\right)^5 \left(\frac{1}{2}\right)^0$

$\Rightarrow 1 \times 0.03125 \times 1$

$P(5) = 0.03125$

No. of families having no girls $= 800 \times 0.03125$

$\Rightarrow \boxed{25}$

(iv) at most 2 girls = at least three boys.

$P(X>3) = P(3) + P(4) + P(5)$

but

$P(0) + P(1) + P(2) + P(3) + P(4) + P(5) = 1$

$P(3) + P(4) + P(5) = 1 - [P(0) + P(1) + P(2)]$

$P(1) = {}^5C_4 \left(\frac{1}{2}\right)^1 \left(\frac{1}{2}\right)^4$

$\Rightarrow 5 \times 0.5 \times 0.0625$

$P(1) = 0.15625$

$P(0) = {}^5C_0 \left(\frac{1}{2}\right)^0 \left(\frac{1}{2}\right)^5$

$\Rightarrow 1 \times 1 \times 0.03125$

$P(0) = 0.03125$

$P(3) + P(4) + P(5) = 1 - [0.03125 + 0.15625 + 0.312]$

$\Rightarrow 0.50$

So, No. of families having at most two girls are.

$= 800 \times 0.50$

$\Rightarrow 400$ families.

(c) If the probability of hitting the target is 10% and 10 shots are fired independently. what is the probability that the target will hit at least once.

Sol $n = 10$

$p = 10\% \Rightarrow 0.1$

$q = 1 - p = 0.9$

$P(X \geq 1) = 1 - P(X = 0)$

$\Rightarrow 1 - {}^{10}C_0 (0.1)^0 (0.9)^{10}$

$P(X \geq 0) = 1 - 3486$

$\boxed{P(X \geq 0) \Rightarrow 0.6513}$

Q13 Suppose that after losing a large amount of money, an unlucky gambler question whether the game was fair and the die was really unbiased - The last 90 tosses of hone the die give the following results.

No. of dots on the dies	1	2	3	4	5	6
No. of times it occurred	20	15	12	17	9	17

Test whether the die is biased or unbiased (given that the tabulated volve of Chi square at 5 degree of freedom at 5% level of significant = 11.07).

Sol Null hypothesis (Ho) = die is biased.

Expected frequency = $\frac{90}{6} \Rightarrow 15$

We have,

$$\chi^2 = \sum \left[\frac{(Oi - Ei)^2}{Ei} \right]$$

~~No. dots on die~~ O_i	20	15	12	17	9	17
E_i	15	15	15	15	15	15
$(O_i - E_i)^2$	25	0	9	4	36	4

$$\chi^2_{cal} = \frac{78}{15} = \boxed{5.2}$$

at 5% significant = 6 − 1 = 5

$$\boxed{\chi^2_{Tab} = 11.07}$$

$$\chi^2_{cal} < \chi^2_{Tab}$$

So, Null hypothesis is accepted

Hence, die is baised.

Q14 Test the hypothesis that there is no difference in the quality of the four kinds of tyres A, B, C and D based on data given below. At 5% level of significant.

	Tyre Brand				Total
	A	B	C	D	
failed to last 40,000 kms	26	23	15	32	96
Lasted from 40000 kms to 60.000 kms	118	93	116	121	448
Lasted more than 60,000 kms	56	84	69	47	256
	200	200	200	200	800

(Given that χ^2_{Tab} value of chi square at 6 degree of freedom at 5% level of significant = 5.348)

Sol Null hypothesis (H_0) = all the quality of tyre are independent.

We have, $$\chi^2_{cal} = \sum\left[\frac{(O_i - E_i)^2}{E_i}\right]$$

Expected frequency

$E(26) = \frac{96 \times 200}{800} = 24$

$E(23) = \frac{96 \times 200}{800} = 24$

$E(15) = \frac{96 \times 200}{800} = 24$

$E(32) = \frac{96 \times 200}{800} = 24$

$E(118) = \frac{448 \times 200}{800} = 112$

$E(93) = \frac{448 \times 200}{800} = 112$

$E(116) = \frac{448 \times 200}{800} = 112$

$E(121) = \frac{448 \times 200}{800} = 112$

$E(56) = \frac{256 \times 200}{800} = 64$

$E(84) = \frac{256 \times 200}{800} = 64$

$E(69) = \frac{256 \times 200}{800} = 64$

$E(47) = \frac{256 \times 200}{800} = 64$

$\Sigma O_i = \Sigma E_i = 800$

O_i	E_i	$(O_i - E_i)^2$	$(O_i - E_i)^2 / E_i$
26	24	4	0.16
23	24	1	0.04
15	24	81	3.37
32	24	64	2.66
118	112	36	0.32.
93	112	361	3.22
116	112	16	0.14
124	112	81	0.72.
56	64	64	1
84	64	400	6.25
69	64	25	0.39
47	64	289	4.51
			22.78

$\chi^2 = 22.78$

at 5% Level of significant degree of freedom =

$(r-1)(c-1) = (3-1)(4-1) \Rightarrow 2 \times 3 \Rightarrow 6$

$\chi^2_{tab} = 5.348$

$\chi^2_{cal} > \chi^2_{tab}$

So, Null hypothesis is rejected.

Hence the quality of tyres are dependent.

[illegible] are females

Q15 The following figures show the distribution of digits in number chosen at random from a telephone directory:

Digits	0	1	2	3	4	5	6	7	8	9	Total.
frequency	1026	1107	997	966	1075	933	1107	972	964	853	10000

Test weather the digits may be taken to occurs equally frequently in the directory.

Sol ⇒ Null hypothesis (H_0) = occurs equally frequently in the directory.

Expected frequency = $\frac{10000}{10} \Rightarrow 1000$

we have

$$\chi^2 = \sum \left[\frac{(O_i - E_i)^2}{E_i}\right]$$

O_i	1026	1107	997	966	1075	933	1107	972	964	853
E_i	1000	1000	1000	1000	1000	1000	1000	1000	1000	1000
$(O_i - E_i)^2$	676	11449	9	1156	5625	4489	11449	784	1296	21609

$\Sigma (O_i - E_i)^2 = 58542$

$\chi^2 = \frac{58542}{1000} \Rightarrow 58.542$

$\boxed{\chi^2_{cal} = 58.542}$

at 5% significant with degree of freedom $\Rightarrow 10 - 1 \Rightarrow 9$

$\chi^2_{Tab} = 16.919$

$\chi^2_{Tal} > \chi^2_{Cal}$

So, Null hypothesis is rejected.

Hence frequently in directory is unequally

Q16 Out of 8,000 graduates in a town 800 are females out of 1600 graduate employs 120 are females, use chi square to determine if any distinction is made in appointment on the basis of sex (given that tabulated value of chi square at 1 degree at 5% left of significant = 3.84)

Solution

	Females	Males	Total
8000 graduates	800	7200	8000
1600 graduates	120	1480	1600
	920	8680	9600

Null hypothesis (Ho) = any distinction is made in appointment is independent

We have $$\chi^2 = \sum\left[\frac{(O_i - E_i)^2}{E_i}\right]$$

Expected frequency.

$$E(800) = \frac{8000 \times 920}{9600} = 766.66$$

$$E(7200) = \frac{8000 \times 8680}{9600} = 7233.33$$

$$E(120) = \frac{1600 \times 920}{9600} = 153.33$$

$$E(1480) = \frac{1600 \times 8680}{9600} = 1446.66.$$

$O_i = E_i = 9600.$

O_i	E_i	$(O_i - E_i)^2$	$(O_i - E_i)^2 / E_i$
800	766.66	1111.55	0.14
7200	7233.33	1110.88	0.15
120	153.33	1110.88	7.24
1480	1446.66	1111.55	0.76
			8.29

$\chi^2_{Cal} = 8.29$

at 5% level of significat at freedom $= (r-1)(c-1) =$
$(2-1)(2-1) = 1$

$\chi^2_{Tab} = 3.84$

$\chi^2_{Cal} > \chi^2_{Tab}$

So, Null hypothesis ~~dis~~ rejected

Hence, any distinctin is made in appointment is dependent.

Q17 Suppose it has been observed that, on average, 180 cars per hour pass a specified point on a particular road in the morning rush hour. Due to impending road works. It is estimated that congestion will occurs closer to the city centre if more than 5 car pass the point in any one minute. what Probability of congestion occurring?

$\mu = P(x=9)$

0 = 0.04979

1 = 0.149361

2 = 0.22404

3 = 0.22404

4 = 0.168031

5 = 0.10082

Sum = 0.91608

P(more than 5 cars arrive in one min) = 1 − P(5 cars or less arrive in one min)

$P(x>5) = 1 - P(x \leq 5)$

$1 - P(x=0) - P(x=1) - P(x=2) - P(x=3) - P(x=4) - P(x=5)$

$P(x=5)$

P(more than 5) = 1 − 0.91608

= 0.08392

P(more than 5) = 0.0839

Q18 The mean number of bacteria per millilitre of a liquid is known to be 6. find the probability that in 1ml liquid there will be : (a) 0 (b) 1 (c) 2 (d) 3 (e) less than 4 (f) 6 bacteria.

Sol

$$P(0) = e^{-6}\frac{6^0}{0!} = 0.00248$$

$$P(x=1) = \frac{\lambda}{1} \times P(x=0) = 6 \times (0.00248)$$

$$= 0.0149$$

$$P(x=2) = \frac{\lambda}{2} \times P(x=1) = \frac{6}{2} \times 0.0149$$

$$= 0.0446$$

$$P(x=3) = \frac{\lambda}{3} \times P(x=2) = \frac{6}{3} \times 0.0446$$

$$= 0.0892$$

$$P(x<4) = P(x=0) + P(x=1) + P(x=2) + P(x=3)$$

$$= 0.1512$$

$$P(x=6) = e^{-6}\frac{6^6}{6!}$$

$$= 0.1606$$

Q19 A Council is considering whether to base a recovery vehicle on a stretch of road to help clear incidents as quickly as possible. The road concerned carries over 5800 vehicles during the peak hour period. Record show that, on average, the number of incidents during the morning rush hour is 5. The council won't base a vehicle on the road if the probability of having more than 5 incidents in any one morning is less than 30%. Based on this information should the council provide a vehicle?

Sol

$P(x > 5)$

$P(x>5) = 1 - P(x \leq 5)$

$P(0) = e^{-5}\frac{5^0}{0!} = 0.00674$

$P(x=1) = 5 \times P(x=0) = 0.03369$

$P(x=2) = \frac{5}{2} \times P(x=1) = 0.08422$

$P(x=3) = \frac{5}{3} \times P(x=2) = 0.14037$

$P(x=4) = \frac{5}{4} \times P(x=3) = 0.17547$

$P(x=5) = \frac{5}{5} \times P(x=4) = 0.17547$

$P(x \leq 5) = P(x=0) + P(x=1) + P(x=2) + P(x=3) + P(x=4) + P(x=5)$

$= 0.61598$

$P(x>5) = 1 - P(x \leq 5)$

$= 0.38403$

$\Rightarrow 38.4\%$